

Q7(a)

For circular motion of the Earth,

Centripetal force = gravitational force exerted by Sun on Earth

$$m r \omega^2 = GMm / r^2$$

$$r^3 = GM \times (T / 2\pi)^2$$

$$r^3 = 6.67 \times 10^{-11} \times 2.0 \times 10^{30} \times (365 \times 24 \times 3600 / 2 \pi)^2$$

$$1 \text{ AU} = r = 1.498 \times 10^{11} \text{ m}$$

$$= 1.50 \times 10^{11} \text{ m}$$

Q7(b)(i)

The mass of the Sun is much bigger compared to the total mass of all the planets, and so their gravitational influence is small.

Q7(b)(ii)

There is no need to expend substantial propellant fuel on the spacecraft to change its momentum.

[OR Gravity assists allow spacecraft to increase their momentum and bend their trajectories from one planet to the next.

OR the spacecraft might not reach a planet such as Saturn before the sun pulls the spacecraft away.]

Q7(b)(iii)

To escape to infinity, Initial KE of object at Jupiter $\geq$ gain in GPE from Jupiter to infinity

$$(1/2)mv^2 \geq 0 - (-GMm / r)$$

$$(1/2) v^2 \geq GM / r$$

$$v^2 \geq (2 \times 6.67 \times 10^{-11} \times 2.0 \times 10^{30}) / (5.2 \times 1.50 \times 10^{11})$$

$$v_{\min} = 1.85 \times 10^4 \text{ m s}^{-1}.$$

Q7(b)(iv)

According to Fig 7.3, the speed of Voyager 2 at Jupiter on 9 July 1979 was $10.5 \text{ km s}^{-1} = 1.05 \times 10^4 \text{ m s}^{-1}$, which is less than the escape speed of $1.85 \times 10^4 \text{ m s}^{-1}$ needed at that distance from the Sun. Hence, it was not travelling fast enough to escape to infinity.

Q7(b)(v)

From Fig. 3, the speed of Voyager 2 increased from 10.5 km s^{-1} to 28.0 km s^{-1} during its interaction with Jupiter.

Hence, its gain in momentum $= \Delta p = p_f - p_i = mv_f - mv_i$

$$= 773 (28.0 - 10.5) \times 10^3$$

$$= 1.35 \times 10^7 \text{ kg m s}^{-1}$$

$$[\text{Accept } 773(28.0 - 10.0) \times 10^3 = 1.39 \times 10^7 \text{ kg m s}^{-1}]$$

Q7(c)(i)

Half-life, $t_{1/2} = 87.74$ years

$$A = A_0 \exp(-\lambda t) \text{ or } A = A_0 \exp(-\ln 2 \times t / t_{1/2})$$

$$A = A_0 \exp(-\ln 2 \times 1/87.74)$$

$$A = A_0 \exp(-0.00790)$$

Activity remaining after one year is $A = 0.992 A_0$

Hence, fractional decrease in activity in one year $= 1 - 0.992 = 0.0079 = 0.79 \%$

Q7(c)(ii)

$$A = 0.623 A_0$$

Power output is proportional to the activity, so

$$P = 0.623 P_0$$

$$P = 0.623 \times 470 = 290 \text{ W}$$

Q7(c)(iii)

Intensity = Power / Area

Power = Intensity $\times$ Area

As the intensity of the received radio waves had decreased, the total area of the receiving parabolic dish antenna needed to be increased so that the total power of the received waves was large enough to be detected.

Q7(c)(iv)

In 1990, distance of Earth from Voyager 1 = 40.47 AU

Angular diameter of Earth = $\theta = d / D$

$$= (2 \times 6.4 \times 10^6) / (40.47 \times 1.50 \times 10^{11}) = \mathbf{2.1 \times 10^{-6} \text{ rad}}$$

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